

Homework 2

1. COMPULSORY EXERCISE

Question 1. At what points is the function continuous?

$$f(x) = \begin{cases} \frac{x^3 - 27}{x^2 - 9}, & x \neq 3, x \neq -3, \\ \frac{9}{2}, & x = 3, \\ 5, & x = -3. \end{cases}$$

Question 2. Using $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, find the following limits. (Here you do **NOT** need to use ε - δ language to prove the results.)

- (a) $\lim_{h \rightarrow 0} \frac{\sin(\sin(2h))}{h}$;
- (b) $\lim_{y \rightarrow 0} \frac{\sin(4y) \cot(7y)}{y \cot(3y)}$;
- (c) $\lim_{x \rightarrow 0} \frac{\tan(2x) - \sin(2x)}{x^3}$;
- (d) $\lim_{x \rightarrow a} \frac{\cos^2 x - \cos^2 a}{x - a}$, where $a \in \mathbb{R}$;
- (e) $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos(3x^2)}}{1 - \cos(2x)}$;

Question 3. Find the limits, then prove your results using the appropriate ε - δ or ε - M definition.

- (a) $\lim_{x \rightarrow -1^+} \frac{x+1}{|x+1|}$;
- (b) $\lim_{x \rightarrow +\infty} \frac{7x^2 + 2}{x^2 + 1}$;
- (c) $\lim_{x \rightarrow +\infty} [\sin(\sqrt{x+2}) - \sin(\sqrt{x-2})]$;
- (d) $\lim_{x \rightarrow a^+} \frac{\sqrt{x} - \sqrt{a} + 2\sqrt{x-a}}{\sqrt{x^2 - a^2}}$, where $a > 0$.

Question 4. Let $\varphi, \psi : \mathbb{R} \rightarrow \mathbb{R}$ be continuous, and define

$$F(x) = \begin{cases} \varphi(x), & x \in \mathbb{Q}, \\ \psi(x), & x \notin \mathbb{Q}. \end{cases}$$

- (a) Prove that F is continuous at a if and only if $\varphi(a) = \psi(a)$.
- (b) Apply (a) to determine all continuity points when $\varphi(x) = \sin x$ and $\psi(x) = -\sin x$.

Question 5. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous at 0 and satisfy $f(x) = f(x^2)$ for every $x \in [0, 1]$.

- (a) Prove that $f(x) = f(0)$ for every $x \in [0, 1]$.
- (b) Must $f(1) = f(0)$? Give a counterexample if not. Prove that the equality does hold if f is also continuous at 1.

Question 6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and define $g(x) = f(x^2)$.

- (a) Using the definitions of limits, prove that, for any $L \in \mathbb{R}$,

$$\lim_{x \rightarrow 0} g(x) = L \iff \lim_{t \rightarrow 0^+} f(t) = L.$$

- (b) If g is continuous at 0, must f be continuous at 0? Prove your answer or give a counterexample.

Question 7. Suppose f, g are defined near a , with $\lim_{x \rightarrow a} f(x) = A$ and $\lim_{x \rightarrow a} g(x) = B$. Define $u(x) = \max\{f(x), g(x)\}$ and $v(x) = \min\{f(x), g(x)\}$.

- (a) Prove directly from the ε - δ definition that $\lim_{x \rightarrow a} u(x) = \max\{A, B\}$ and $\lim_{x \rightarrow a} v(x) = \min\{A, B\}$.
- (b) Conversely, if both u and v have finite limits at a , must both f and g have limits there? Give a proof or a counterexample.

Question 8. Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at $x = 0$. If $f(x + y) = f(x) + f(y)$ for every $x, y \in \mathbb{R}$, prove that f is continuous at every point of $\mathbb{R}$.

2. EXTRA EXERCISES

Question 9. Consider functions $f : [0, +\infty) \rightarrow \mathbb{R}$.

- (a) Suppose $f(x + 1) = f(x)$ for every $x \geq 0$ and $\lim_{x \rightarrow +\infty} f(x) = L \in \mathbb{R}$. Prove that $f(x) = L$ for all $x \geq 0$.
- (b) Suppose instead that f is bounded and continuous, and $\lim_{x \rightarrow +\infty} (f(x + 1) - f(x)) = 0$. Must f have a limit at $+\infty$? If not, construct a counterexample and verify all its required properties.

Question 10. Suppose that $f : (0, 1) \rightarrow \mathbb{R}$ satisfies

$$\lim_{x \rightarrow 0^+} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^+} \frac{f(2x) - f(x)}{x} = 0.$$

Prove that

$$\lim_{x \rightarrow 0^+} \frac{f(x)}{x} = 0.$$

Hint: If $0 < x < \delta$, then $0 < x/2^n < \delta$ for every integer $n \geq 1$. You may use $\sum_{n=1}^{\infty} 2^{-n} = 1$ and $\lim_{n \rightarrow \infty} f(x/2^n) = 0$ directly for each fixed $x \in (0, 1)$.

3. THOMAS TEXTBOOK EXERCISES

Chapter 1. Limits and Differentiation**Section 1.1. Limits and continuity**

§1.1.4. One-sided limits

Exercises 2.4. 20, 32, 52.

§1.1.5. Continuity

Exercises 2.5. 30, 38.

§1.1.6. Limits involving infinity, asymptotes of graphs

Exercises 2.6. 36, 52, 84, 102.*Chapter 2 Additional and Advanced Exercises.* 32, 34.

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